

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

Lecture 2: Measures of Model Performance – Supervised Model Introduction

Measure of Model Performance: Topics

- Linear algebra review
- Supervised model introduction
- Metrics of performance
- Be able to create receiver operating characteristic curves
- kNN – a simple supervised learning model

Making prediction using a trained model - Regression

- Linear Regression model:

$$model(\mathbf{x}, \mathbf{w}^*) = \mathbf{x}^T \mathbf{w}^* = w_0^* + x_1 w_1^* + \cdots + x_N w_N^*$$

$\mathbf{w}^*$: the optimal set of weights found by minimizing a regression cost function.
Prediction: Given input $\mathbf{x}$, the output $\hat{y}$ is predicted by

$$model(\mathbf{x}, \mathbf{w}^*) = \hat{y}$$

Judging the quality of a trained regression model (1)

- Least Squares: Mean Square Error (MSE)

$$MSE = \frac{1}{N} \sum_{i=1}^N (model(\mathbf{X}_i, \mathbf{w}^*) - \mathbf{y}_i)^2$$

- Least Absolute Deviation: Mean Absolute Deviation (MAD):

$$MAD = \frac{1}{N} \sum_{i=1}^N |model(\mathbf{X}_i, \mathbf{w}^*) - \mathbf{y}_i|$$

Judging the quality of a trained regression model (2)

- Least Squares: Mean Square Error (MSE)

$$MSE = \frac{1}{N} \sum_{i=1}^N (model(\mathbf{x}_i, \mathbf{w}^*) - \mathbf{y}_i)^2$$

- Least Absolute Deviation: Mean Absolute Deviation (MAD):

$$MAD = \frac{1}{N} \sum_{i=1}^N |model(\mathbf{x}_i, \mathbf{w}^*) - \mathbf{y}_i|$$

- Comparison
 - MSE is sensitive to outliers
 - MAD is better measure of error than MSE if forecast error does not have asymmetric distribution

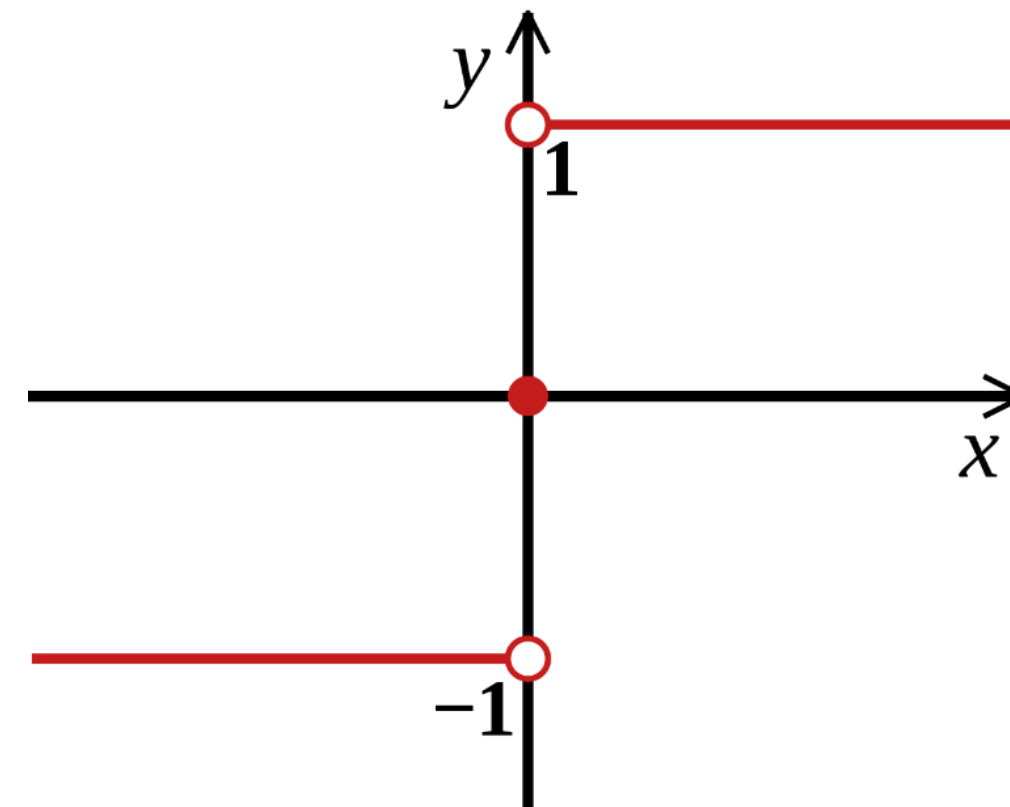
Making predictions using a trained model – Classification (1)

- Classification model (PLA):

$$model(\mathbf{x}, \mathbf{w}^*) = sign(\mathbf{x}^T \mathbf{w}^*) = sign(\mathbf{w}_0^* + x_1 \mathbf{w}_1^* + \cdots + x_N \mathbf{w}_N^*)$$

- Labels: $y_p = \{-1, +1\}$

- $sign(x) = \begin{cases} +1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$



Making predictions using a trained model – Classification (2)

- Classification model (PLA):

$$model(\mathbf{x}, \mathbf{w}^*) = sign(\mathbf{x}^T \mathbf{w}^*) = sign(w_0^* + x_1 w_1^* + \cdots + x_N w_N^*)$$

Prediction: Given input $\mathbf{x}$, the output y is predicted by

$$model(\mathbf{x}, \mathbf{w}^*) = \hat{y}$$

Judging the quality of a trained model using accuracy (1)

- The prediction of $\mathbf{x}_i$ is $\hat{\mathbf{y}}_i = \text{model}(\mathbf{x}_i, \mathbf{w}^*)$, and the true label is $\mathbf{y}_i$.
- Indicator function: returns 1 where condition is true

$$I_{[\hat{\mathbf{y}}_i = \mathbf{y}_i]} = \begin{cases} 0 & \text{if } \hat{\mathbf{y}}_i \neq \mathbf{y}_i \\ 1 & \text{if } \hat{\mathbf{y}}_i = \mathbf{y}_i \end{cases}$$

- Number of correctly labeled samples: $\sum_{i=1}^N I_{[\hat{\mathbf{y}}_i = \mathbf{y}_i]}$
- Accuracy: $A = \frac{1}{N} \sum_{i=1}^N I_{[\hat{\mathbf{y}}_i = \mathbf{y}_i]}$
 - The accuracy ranges from 0 (no points are classified correctly) to 1 (all points are classified correctly)

Judging the quality of a trained model using balanced accuracy (2)

- Classification accuracy is easy and excellent, but it can paint an incomplete picture of how well we have really solved a classification problem

Judging the quality of a trained model using balanced accuracy (3)

- Classification accuracy is easy and excellent, but it can paint an incomplete picture of how well we have really solved a classification problem
 - A dataset consists of highly imbalanced classes: if one class makes up 95% of all data points, a naïve classifier that blindly assigns the label of the majority class to every training point achieves an accuracy of 95% but here misclassifying 5% amounts to completely misclassifying an entire class of data.

Judging the quality of a trained model using balanced accuracy (4)

- Balanced Accuracy

- Denote the indices of samples with labels $y_i = +1$ and $y_i = -1$ as Ω_{+1} and Ω_{-1}

- Number of correctly classified samples on $+1$ class: $\sum_{i \in \Omega_{+1}} I[\hat{y}_i = y_i]$

- Number of correctly classified samples on -1 class: $\sum_{i \in \Omega_{-1}} I[\hat{y}_i = y_i]$

- Accuracy:

- $A_{+1} = \frac{1}{|\Omega_{+1}|} \sum_{i \in \Omega_{+1}} I[\hat{y}_i = y_i]$

- $A_{-1} = \frac{1}{|\Omega_{-1}|} \sum_{i \in \Omega_{-1}} I[\hat{y}_i = y_i]$

- $A_{balanced} = \frac{A_{+1} + A_{-1}}{2}$

Confusion Matrix and Accuracy (1)

		Predicted	
		Positive	Negative
Actual	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

- Accuracy: $A = \frac{TP+TN}{TP+TN+FP+FN}$
- Accuracy of each class: $A_+ = \frac{TP}{TP+FN} = TPR$, $A_- = \frac{TN}{TN+FP} = TNR$
- Balanced Accuracy: $\frac{1}{2}A_+ + \frac{1}{2}A_- = \frac{1}{2}TPR + \frac{1}{2}TNR = \frac{1}{2}\frac{TP}{TP+FN} + \frac{1}{2}\frac{TN}{TN+FP}$

Confusion Matrix and Accuracy (2)

		Predicted	
		Positive	Negative
Actual	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$
- $F_1: \frac{2TP}{2TP+FP+FN} = \frac{TP}{TP+\frac{1}{2}(FP+FN)}$

Confusion Matrix and Accuracy (3)

		Predicted	
		Positive	Negative
Actual	Total Population	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$
- $F_1 = \frac{2TP}{2TP+FP+FN} = \frac{TP}{TP+\frac{1}{2}(FP+FN)}$
- Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$
- Balanced Accuracy = $\frac{1}{2} \frac{TP}{TP+FN} + \frac{1}{2} \frac{TN}{TN+FP}$

Example (1)

		Predicted	
		Positive	Negative
Actual	Positive	TP=25	FN=5
	Negative	FP=5	TN=25

- $\text{Precision} = \frac{TP}{TP+FP}$
- $\text{Recall} = \frac{TP}{TP+FN}$
- $F_1 = \frac{2TP}{2TP+FP+FN} = \frac{TP}{TP+\frac{1}{2}(FP+FN)}$
- $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$
- $\text{Balanced Accuracy} = \frac{1}{2} \frac{TP}{TP+FN} + \frac{1}{2} \frac{TN}{TN+FP}$

Example (2)

		Predicted	
		Positive	Negative
Actual	Positive	TP=25	FN=5
	Negative	FP=5	TN=25

- Precision = $\frac{TP}{TP+FP} = \frac{25}{25+5} = 0.833$
- Recall = $\frac{TP}{TP+FN} = \frac{25}{25+5} = 0.833$
- $F_1 = \frac{2TP}{2TP+FP+FN} = \frac{TP}{TP+\frac{1}{2}(FP+FN)} = \frac{25}{25+\frac{1}{2}(5+5)} = 0.833$
- Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} = \frac{25+25}{25+25+5+5} = 0.833$
- Balanced Accuracy = $\frac{1}{2} \frac{TP}{TP+FN} + \frac{1}{2} \frac{TN}{TN+FP} = \frac{1}{2} \frac{25}{25+5} + \frac{1}{2} \frac{5}{25+5} = 0.5$

Practice 1

		Predicted	
		Positive	Negative
Actual	Positive	TP=2	FN=3
	Negative	FP=85	TN=201

- Calculate the accuracy based on the given confusion matrix

Practice 2

		Predicted	
		Positive	Negative
Actual	Positive	TP=2	FN=3
	Negative	FP=85	TN=201

- Calculate the precision, recall, and F_1 score based on the given confusion matrix

F-measure (1)

Why F-measure?

		Predicted	
		Positive	Negative
Actual	Total Population		
	Positive	TP=2	FN=3
	Negative	FP=85	TN=201

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN} = \frac{2+201}{2+85+3+201} = 0.6975945017$
- Not bad if only looking at accuracy.
- If only looking at class +1, this is a poor classifier. Most samples are classified to class -1.

F-measure (2)

Why F-measure?

		Predicted	
		Positive	Negative
Actual	Total Population		
	Positive	TP=2	FN=3
	Negative	FP=85	TN=201

- $Accuracy = \frac{TP+TN}{TP+TN+FP+FN} = \frac{2+201}{2+85+3+201} = 0.6975945017$
- $Precision = \frac{TP}{TP+FP} = \frac{2}{2+85} = 0.0229885057$
- $Recall = \frac{TP}{TP+FN} = \frac{2}{2+3} = 0.4$
- $F1 = \frac{TP}{TP + \frac{1}{2}(FP+FN)} = \frac{2}{2 + \frac{1}{2}(85+3)} = 0.0434782609$

Measure of Model Performance: Topics (End)

- Linear algebra review
- Supervised model introduction
- Metrics of performance
- Be able to create receiver operating characteristic curves
- kNN – a simple supervised learning model